SUDHANTHIRAPRIYA SUNDARAM

Machine Learning Engineer | AI / LLM Engineering

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 2 years of experience (internships + current role) shipping production AI systems across RAG-based regulatory assistants, ERP recommendation services, and bilingual (Tamil/English) OCR pipelines processing government records at ~98% accuracy. Experienced across the full pipeline: data/web scraping, OCR, vector search & hybrid retrieval (Qdrant), ML classification, and Flask/MCP API delivery. Looking to deepen cloud deployment and MLOps skills in a next role.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | *AK Consultants* Mar 2025 – Present

- Own end-to-end data pipelines and ML delivery across multiple production AI systems, spanning data collection, ML modeling, retrieval, and API development.

AI-Assisted UK Road & Transport Planning Platform (Backend & Chatbot, Confidential Client)

- Built and deployed Flask API endpoints for a Plotly-based chart-generation module, turning traffic and footfall CSV data into on-demand analytics consumed directly by the frontend.
- Designed and trained a standalone ML classifier to flag uploaded PDFs as relevant or irrelevant to UK transport standards before ingestion into the platform's RAG pipeline (FAISS + HuggingFace embeddings + Groq/Gemini LLMs).
- Built web scraping and PDF-scraping pipelines to source and structure UK transport guideline documents (Highway Code, Manual for Streets, TSRGD).
- Contributed to the platform's LLM-powered chatbot, letting users query transport data and regulatory documents in natural language via the RAG pipeline, making information retrieval easier for non-technical users.

LLM-Powered PDF-to-Excel Extraction Tool (MCP, 2-person team)

- Co-built a privacy-first PDF extraction tool combining LLM-based extraction (self-hosted Ollama) with rule-based table extraction (Camelot/Tabula) for complex and tabular PDF layouts.
- Set up and integrated local LLM inference end-to-end via Ollama, removing dependence on cloud AI services for sensitive documents (e.g., property/HMO license records).
- Currently repackaging the tool as an MCP (Model Context Protocol) server, pre-production, for adoption by a colleague team as a callable extraction tool.
- Implemented dynamic column detection, reusable extraction templates, and multi-level data validation, exposed via a Streamlit web UI and a CLI batch pipeline.

Course Curator — AI Regulatory Intelligence & Recommendation Platform (Indian Higher Education) — Live

- Built the Resource and Trend scoring services powering the platform's curriculum recommendation engine for Indian higher-education institutions (AICTE/UGC/PCI/COA-aligned).
- Ran iterative chunking and ingestion experiments into Qdrant across large regulatory document collections, tuning strategy to improve retrieval quality within the platform's hybrid vector + graph retrieval pipeline.
- Collaborated with the team on deployment and staging environment setup on Google Cloud Platform (Cloud Run).

Urimai — AI-Driven Public Election Affidavit Transparency Platform — Live

- Built a bilingual (Tamil + English) OCR extraction pipeline for scanned candidate affidavits, evaluating EasyOCR, Tesseract, and a custom fine-tuned model before converging on a Google Cloud Vision + Gemini hybrid, reaching ~98% production extraction accuracy.
- Engineered LLM-based structuring of OCR output into 33 validated fields per affidavit (personal, financial, legal data) and built the JSON-to-Excel and ETL pipelines consolidating multi-year candidate records.
- Designed a unique candidate-ID system for cross-year person matching and deduplication, combining normalized names, constituency/year mapping, and Google Translate API for Tamil↔English transliteration.

AI Engineer Intern | *Digital Garage Pvt. Ltd.* Mar 2024 – Feb 2025

- Built and optimized ML models using Python and TensorFlow, iterating on architecture and hyperparameters to improve performance.
- Collaborated on AI-based problem-solving initiatives and evaluated model performance against project requirements.

Machine Learning Intern | *CubenSquare* May 2023 – Jul 2023

- Applied transfer learning techniques to pre-trained models to improve outcomes on limited data and reduce training time versus training from scratch.

ACADEMIC & KAGGLE PROJECTS

Text-to-Image Generation Model — Stable Diffusion model for high-fidelity image generation from text prompts. Demo
Multiple Eye Disease Detection — CNN-based model to predict eye conditions from mobile camera images.
Employee Attrition Prediction — Decision Tree model predicting attrition to support retention planning. Demo
Smartphone Addiction Prediction (Kaggle) — ROC-AUC 0.966 using a tuned 3-model gradient boosting ensemble (XGBoost, LightGBM, CatBoost); ranked 693/3,561 (top ~19%).
Student Health Risk Prediction (Kaggle) — ML model predicting student health risk from survey and lifestyle features.
House Price Prediction (Kaggle) — Regression model predicting housing prices from property features. Kaggle Profile

TECHNICAL SKILLS

Programming: Python, SQL, HTML, CSS

LLMs & GenAI: RAG Pipelines, Local LLM Deployment (Ollama), MCP (Model Context Protocol), Groq/Gemini APIs, Prompt Engineering, LLM Evaluation

Machine Learning: Classification (KNN & custom classifiers), Deep Learning, NLP, Feature Engineering, Model Evaluation

Backend & APIs: Flask REST APIs, PostgreSQL, Redis, FAISS & Qdrant Vector Search, HuggingFace Embeddings

Data & Visualization: Data Preprocessing, Web Scraping, PDF/Table Extraction (Camelot, Tabula), OCR, Plotly Chart Generation

Libraries & Tools: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Streamlit

Soft Skills: Problem Solving, Communication, Teamwork, Time Management, Adaptability

EDUCATION

B.Tech in Artificial Intelligence and Data Science — Arjun College of Technology | CGPA: 8.3 | Jul 2020 – May 2024
Higher Secondary Certificate (HSC) — M.C.T Higher Secondary School | 70% | Jan 2019 – Mar 2020

CERTIFICATIONS

Designing Scalable RAG Pipelines — Scaler · Building LLM Applications with Prompt Engineering — NVIDIA · Introduction to Machine Learning — Kaggle · Introduction to Programming Using Python — Kaggle · Data Science with Python — Great Learning · Machine Learning in Python — Edureka · Cloud Computing — NPTEL · Python Full Stack — Shiash Info Solutions · Linux Administration — Arjun College of Technology

LANGUAGES

Tamil, English